ISYS 1108 / 3474 Software Engineering Project Management

Dr Maria Spichkova

Senior Lecturer in Software Engineering

Week 2: Lectorial

What's next...

RMIT UNIVERSITY

Acknowledgement of Country

RMIT University acknowledges the people of the Woi wurrung and Boon wurrung language groups of the eastern Kulin Nation on whose unceeded lands we conduct the business of the University.

RMIT University respectfully acknowledges their Ancestors and Elders, past and present.

RMIT also acknowledges the Traditional Custodians and their Ancestors of the lands and waters across Australia where we conduct our business.

Classes and Deadlines

Mid-semester break is between Weeks 5 and 6

	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12
Lectorials												
Workshops						Team-based assignment activities						
A1: SE Challenge				Friday: Challenge 1 10%					Friday: Challenge 2 10%		Friday: Challenge 3 10%	
A2: Team-based assignment								Due on Friday: 30%				
A3: Individual Assignment												Due on Friday: 40%

Our plan for this session

- 😊 Agile
- 😊 Scrum
- 😊 Product backlog
- 😊 User stories

Agile

The Agile Scrum Framework at a Glance
AGILE
Agile
AGILE DEVELOPMENT
YES
NO
NEXT ITERATION
Agile
Final Sprint
LAUNCH
Agile Development
Agile Methodology
Sprint 1
Sprint 2
Sprint 3
AGILE MANIFESTO
Benefits of AGILE
AGILE
Scrum
team
product
backlog
sprint
items
work
Product
incrementally
instead of all at once
EVALUATE
SCOPE
THE AGILE METHODOLOGY
DEVELOP
DEFINE
MODERN agile
VALUES

Agile

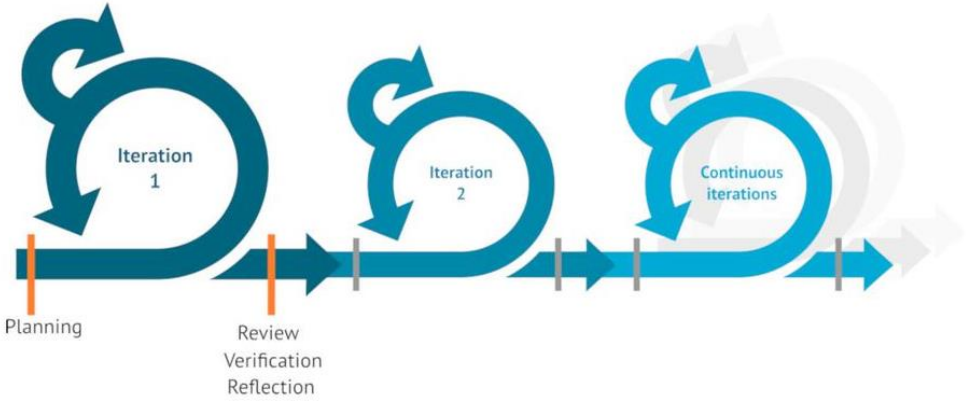
A framework for delivering products quickly and efficiently.

It encompasses set of exciting new practices and techniques that make product development more cyclical (or ***iterative***).

It relies on **lean governance (management)** as opposed to more traditional techniques that rely on **heavyweight governance**.

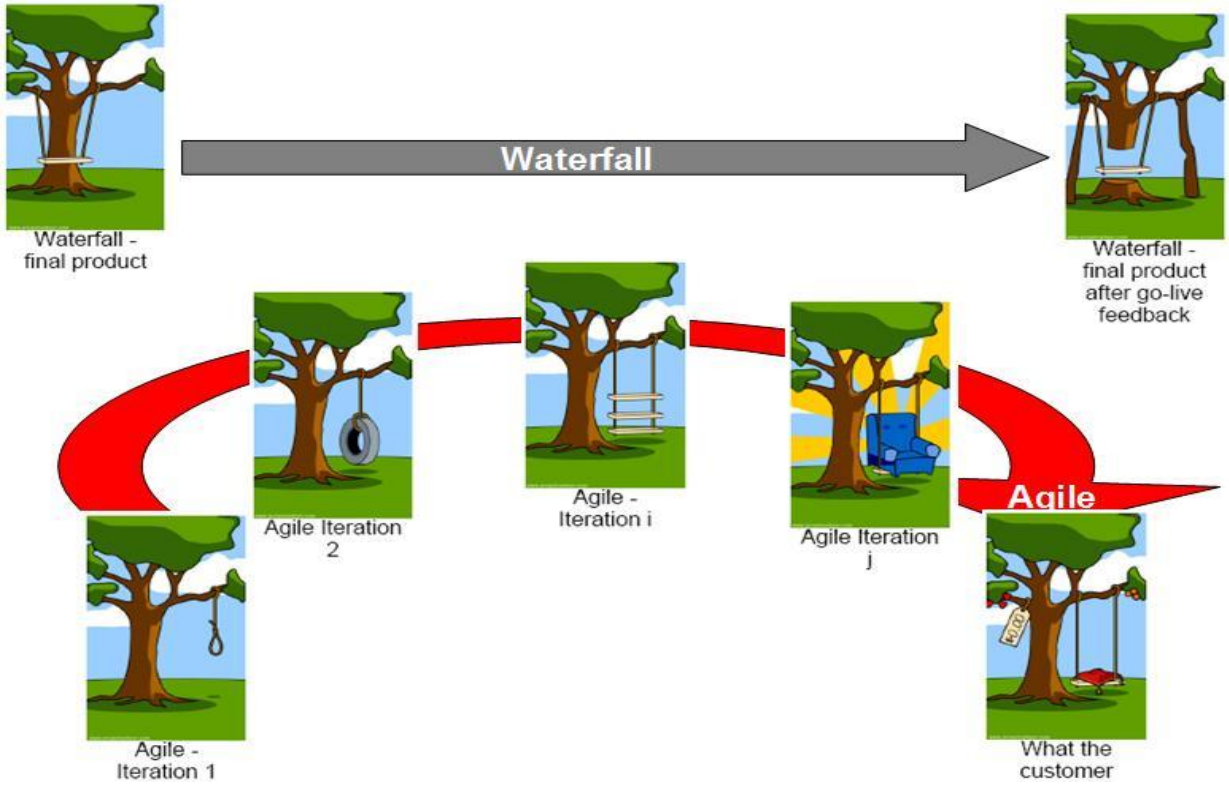
Agile is all about empowering the team and getting closer to what the customer wants.

As opposed to the standard requirements-gathering, plan-driven project development, the customer is continually involved in Agile.



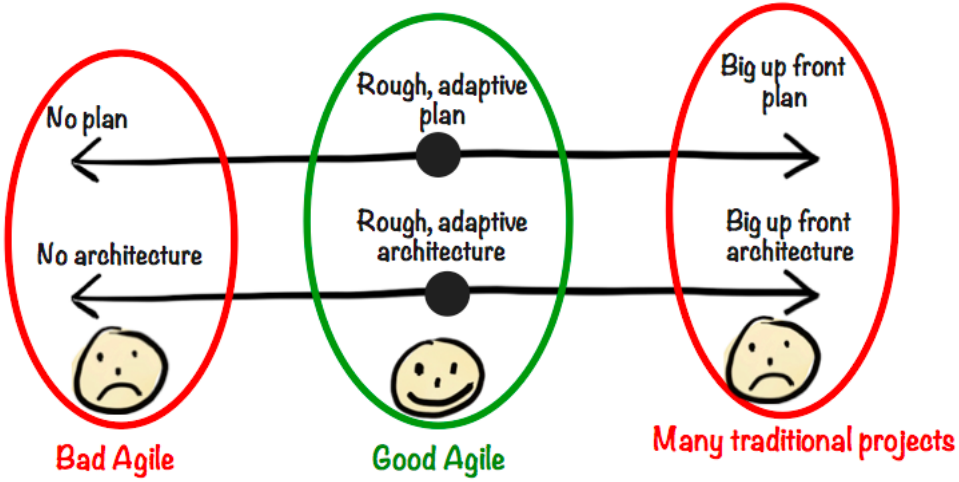
Why Agile was created?

It was a reaction against the rigid and “big upfront planning” approach that many software developers and business analysts thought stifled the development process and contact with the customer.



Agile wasn’t a simple software developer’s revolt. While Agile became popular with software developers because it freed time for creativity and drove the decision-making process lower, the principles also made sense from a business perspective.

Good Agile


No plan
Rough, adaptive plan
Big up front plan
No architecture
Rough, adaptive architecture
Big up front architecture
Bad Agile
Good Agile
Many traditional projects

The principles of Agile were in the air for a long time when they were finally put to paper in the **Manifesto for Agile Software Development** (aka, **Agile Manifesto**) in 2001 by its 17 cosigners at The Lodge at the Snowbird ski resort in the Wasatch Range of mountains in Utah (USA).

Agile vs. Scrum

Scrum is a particular variant of Agile and the most popular variant today.

While Agile is more a set of principles, Scrum puts principles into effect with well-designed practices and techniques:

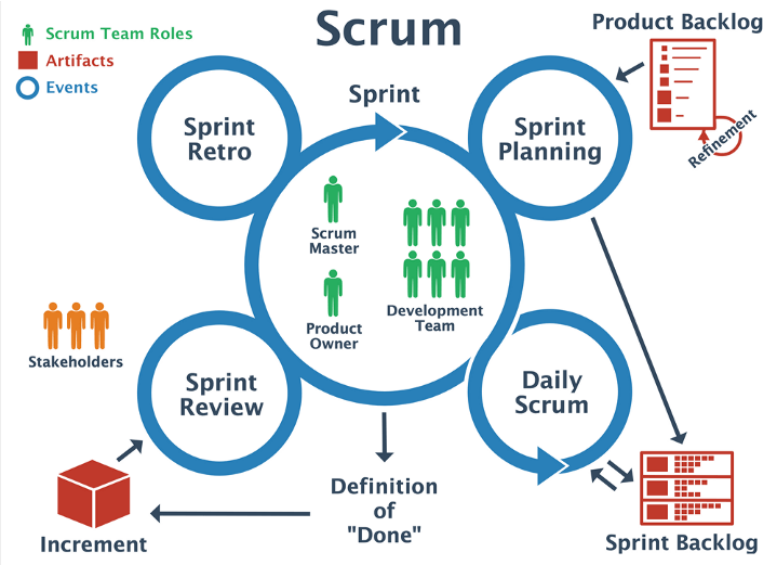
Agile may give you the basic orientation, but Scrum lets you bring Agile to your organisation and make it work.

While Agile suggests a need for good communication, Scrum sets up rules for stand-up meetings and how they should be conducted.

Where Agile discusses the need for iterative and incremental development, Scrum brings those terms into focus with well-defined sprints that represent a specific iteration.

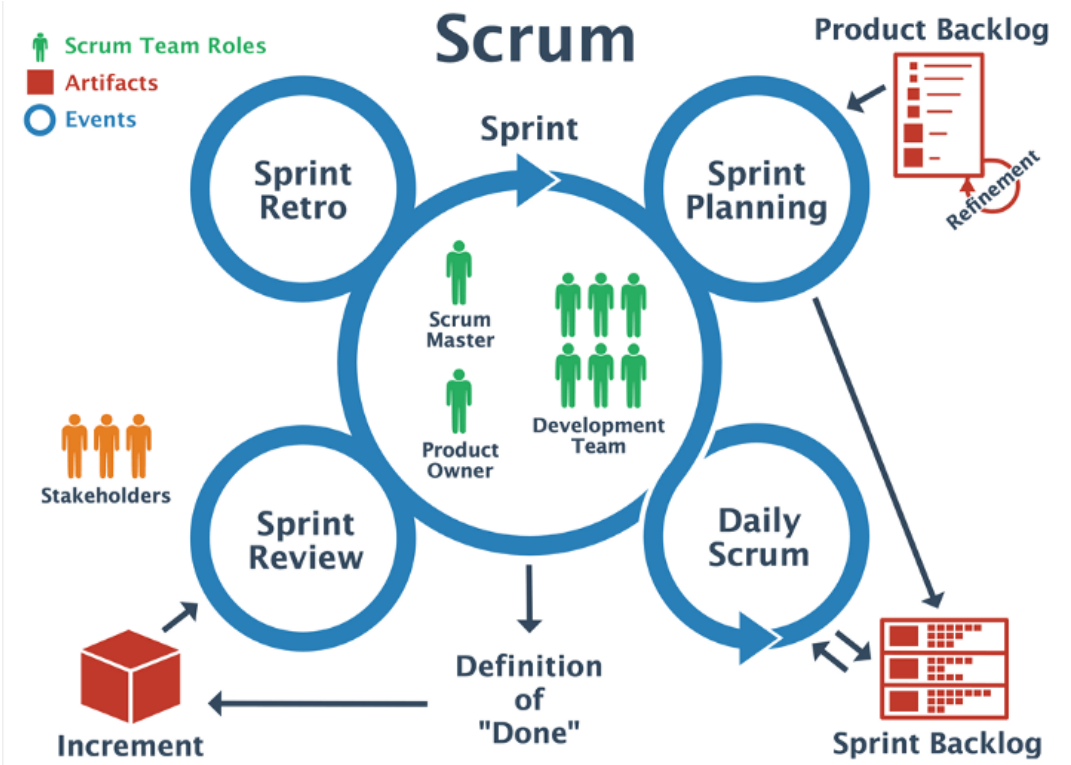
Where Agile suggests you track tasks to be performed with lists, Scrum has specific techniques in place for doing exactly that.

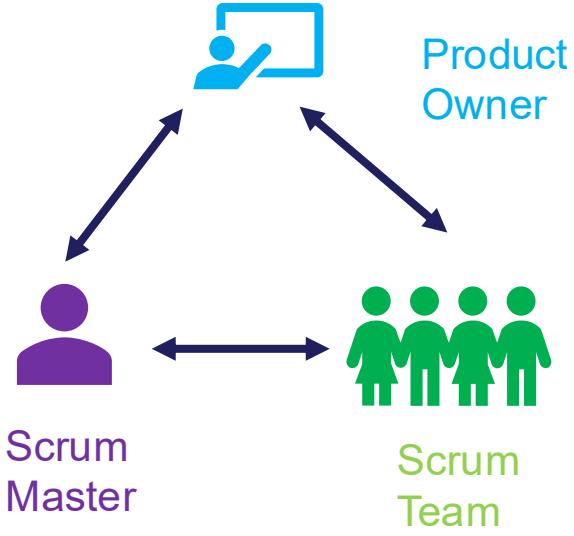
Scrum


Scrum
Scrum Team Roles
Artifacts
Events
Sprint
Sprint Retro
Sprint Planning
Product Backlog
Refinement
Scrum Master
Product Owner
Development Team
Stakeholders
Sprint Review
Daily Scrum
Increment
Definition of "Done"
Sprint Backlog

Scrum

- team-based iterative and incremental Agile methodology.
- has its own components, such as the Scrum Team, backlogs, and sprints *etc.*





Product owner

- Establishes and communicates the product vision
- Achieves initial and on-going funding for the project
- Creates initial release plans
- Makes decision about when to create an **official release:** it may not be desirable to release at the conclusion of every increment
- Develops and maintains/prioritizes the **Product Backlog**, as well as presents and explains it to the team
- Empowered to make decisions for all customers and users
- May attend sprint planning meeting and sprint review meeting

Voice of the customer

Owns value

Gathers feedback

Makes decisions

Scrum Master

- Responsible for the process
- Responsible for maximising team productivity and removing obstacles to progress
- Sets up and conducts meetings:
  - Sprint Planning Meetings, during which the team expands Product Backlog into Sprint Backlog
  - Sprint Review Meetings, during which the newly developed functionality is demonstrated
  - Short daily stand-ups
  - …
- Representative to management
- Representative to team

Scrum Team

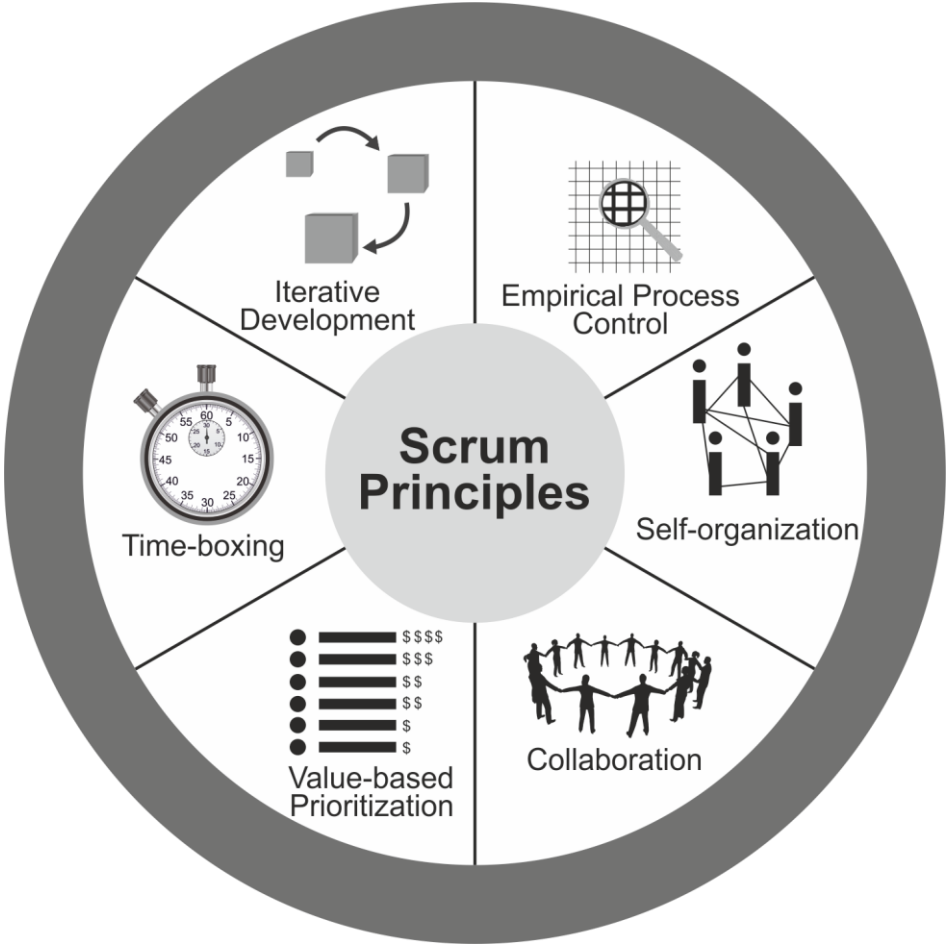
During an iteration, the team selects and develops the highest-priority features on the Product Backlog

The team expands Product Backlog Items into more explicit tasks on a **Sprint Backlog** and then manages its own work and self-organises around how it desires to complete the iteration.

- ✓ Typically 6 - 12 people
- ✓ Cross-functional to all the necessary tasks to be handled (for example, including programmers, architect, designers, QA/testers, tech. writer, …)
- ✓ **Self-organised and self-directed**
- ✓ Owns the project

It will only work if the team members are

- ✓ Experienced
- ✓ Motivated
- ✓ Committed to the project
- ✓ Proud of their work
- ✓ Able to work well with others
- ✓ Responsible
- ✓ **Willing to work in teams**
- ✓ **Autonomous**


Scrum Principles
Iterative Development
Empirical Process Control
Self-organization
Collaboration
Value-based Prioritization
Time-boxing

Questions?

Let’s go to www.menti.com ☺

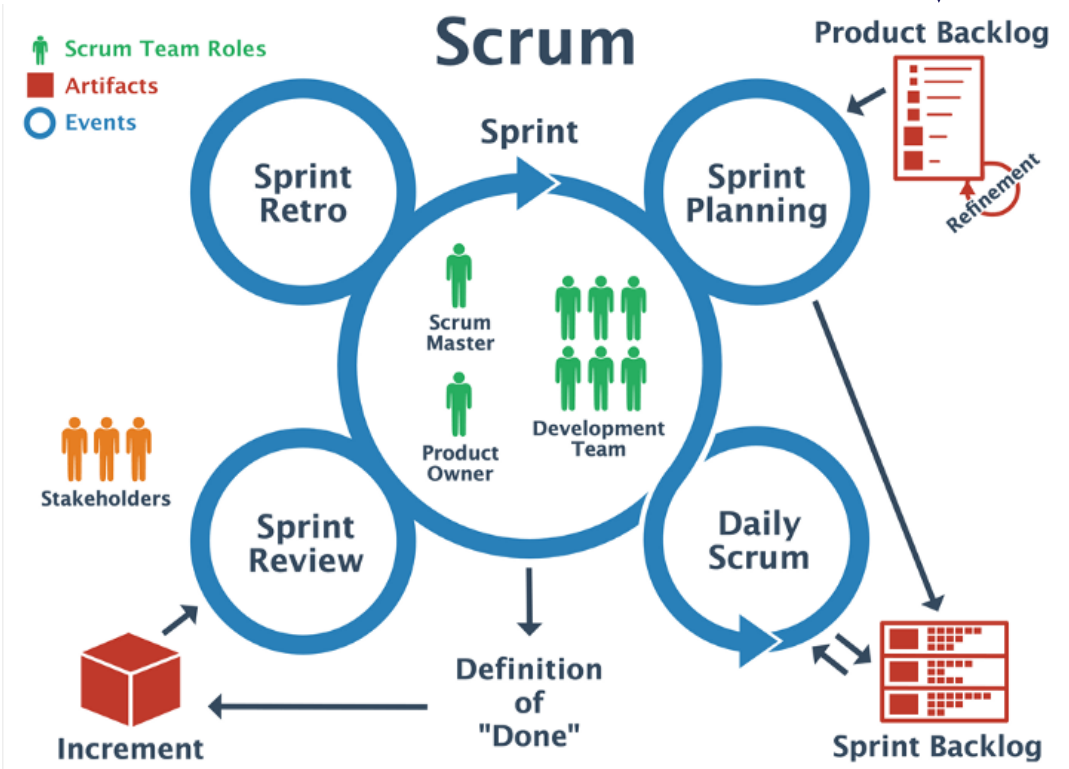
PRODUCT BACKLOG: Project Name

ID	USER STORY	OWNER	PRIORITY	EFFORT	STATUS	DEFINITION OF DONE	NOTES

Product backlog

Scrum

Vision & Scope Document / Project Charter



- We will discuss all these artefacts within this course.
- The first one we have to create (on the basis of Vision & Scope document or Project Charter) is Product backlog. – Let's start with it!

Project Management Artefacts in Scrum

Project Charter
or Vision & Scope Document

Product Backlog

Sprint 1
planning notes

Sprint Backlog (Sprint 1)

Sprint 1
retro notes

Sprint 2
planning notes

Sprint Backlog (Sprint 2)

Sprint 2
retro notes

Sprint 3
planning notes

Sprint Backlog (Sprint 3)

Sprint 3 retro
notes

...

Burn-
down
chart

Sprint 0

Sprints
1, 2, 3,
4, 5, …

Product Backlog

- Requirements for a system, expressed as a prioritized list
- Each item should have value to the users/ customers
- Is managed and owned by a Product Owner
- Spreadsheet, Trello board, etc.

Most **Product Backlog Items (PBIs)** initially need refinement because they are too large and poorly understood / vague / abstract.

PRODUCT BACKLOG: Project Name

ID	USER STORY	OWNER	PRIORITY	EFFORT	STATUS	DEFINITION OF DONE	NOTES

Can be changed and re-prioritized at the start of each sprint, before the Sprint Planning Meeting

Product Backlog (in table format)

IDs should be unique and meaningful:
In simplest case, we can use just numbers. Typically, also some prefixes are used in addition to numbers

ID	USER STORY	OWNER	PRIORITY	EFFORT in story points	STATUS	DEFINITION OF DONE	NOTES
PBI1							
PBI2							
PBI3							
PBI4							
PBI5							
PBI6							
PBI7							
PBI8							
PBI9							
…							

For all functional items (== items describing functionality) the owner is Product Owner

Priority:
- depends on importance of the item (decided by Product Owner)
- Typically denoted as Critical/ High/ Medium / Low

Status is typically denoted as ToDo / In Progress / Done

Product Backlog (in table format)

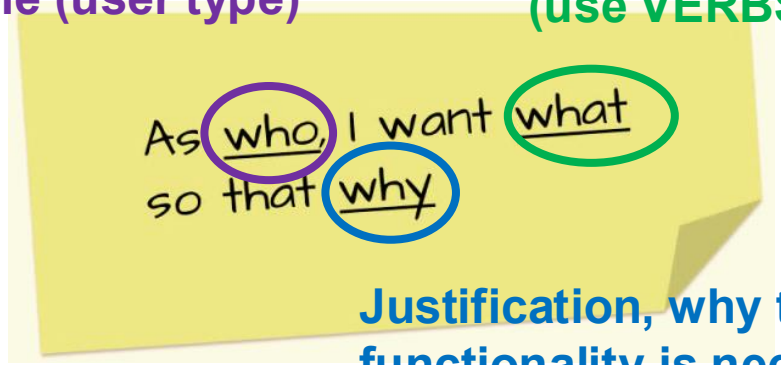
ID	USER STORY	OWNER	PRIORITY	EFFORT in story points	STATUS	DEFINITION OF DONE	NOTES
PBI1							
PBI2							
PBI3							
PBI4							
PBI5							
PBI6							
PBI7							
PBI8							
PBI9							
…							

Let’s now discuss what do we mean under user stories, efforts, and Definition of Done!

User story

Role (user type)

Activity to describe the required functionality (use VERBS!)



Justification, why this functionality is needed

RE in Agile projects: User stories

Agile:

- Lightweight approach to requirements engineering, project management and other aspects of software development
- Business Analyst is often the Developer ☺

User stories:

- Less detailed that “classical requirements specification”
- But still be written and provide some product documentation
- Seek to combine the strengths of written and verbal communication, where possible supported by a picture.
- **Written in user language**

As a [user role]
I want to [goal action]
So that [reason/benefit]

Example:
As a registered user, I want to change my password so that I can keep it secure or make it easier to remember.

User Stories

User stories are very flexible and can range from individual functional requirements, scenarios, use cases, product features, business objectives, constraints, quality attributes, business rules, user interface issues, and desired characteristics of the product

- Deliberately leave out a lot of important details: Help getting the high-level picture of requirements (provide a “bone” for eliciting during scrum meetings)
- Product Backlog is evolving during the whole length of the project.
- Elaborated and modified through the project based on user input
- Centered on the result and the benefit of the thing you're describing
- **User- and usage-centric requirements elicitation:**
  focus on what the user wants to do, and not what the user wants the system to do:
  By focusing on the user’s goal and vision, you can ensure that the users get the functionality that they require

Negotiable
Independent
Valuable
Estimable
Small
Testable

User stories: Core characteristics

- Are **not a contract**, not detailed specifications.
- Are a kind of **"notes"** to remind the team on to discuss and collaborate to clarify the details near the time of development
- High level, no details, as the goal is
  - To avoid a false impressions of precision or completeness
  - To highlight that further discussions are a part of the process
- Should be **as independent as possible**
- Dependencies might lead to difficulties with prioritizing and planning
- Should be **valuable to the user** (or owner) of the product – like traditional req. specification should contain only necessary requirements

Negotiable

Independent

Valuable

User stories: Core characteristics

- Need to provide enough information to **estimate**, without being too detailed.
- Should be "**small enough**": small in the sense of the feature scope, not in the sense "very short text"
- Should be detailed enough for the team to start work from, and further details can be established and clarified at the time of development.
- Generally: A team must be able to finish a user story **within the sprint** in which it starts
- Should provide enough details of how the User Story will be **tested**.

Estimable

Small

Testable

Is this story negotiable?

As an HR staff member, I want to pay for job posting with a credit card so that I can have efficient payment process.

Note: Accept Visa, MasterCard, and American Express. Consider Discover. On purchases over $100, ask for card ID number from back of card. The system can tell what type of card it is from the first two digits of the card number. The system can store a card number for future use. Collect the expiration month and date of the card.

How about this one?

As an HR staff member, I want to pay for job posting with a credit card so that I can have efficient payment process.
Note: Will we accept Discover cards?
Note for UI: Don't have a field for card type (it can be derived from first two digits on the card).

Which users should you target while writing user stories?

Broaden the scope from looking at one user

Allows users to vary by

- What they use the software for
- How they use the software
- Background
- Familiarity with the software / computers

Definition:

A **user role** is a collection of defining attributes that characterize a population of users and their intended interactions with the system.

Which variant is correct?

Assume we need to specify a functionality "cancellation of hotel reservation", which should be available for all users of a booking site

1) **As a** user, **I want** to have a cancellation functionality.

2) **As a** user, **I want** to have a cancellation functionality, **so that** I can cancel my booking.

3) **As a** user, **I want** to have a cancellation functionality, **so that** I can save on cancellation penalty.

4) **As a** user, **I want** to have a "cancel booking" button, **so that** I can cancel my booking.

5) **As a** user, **I want** to be able to cancel my hotel reservations, **so that** I can save on cancellation penalty.

6) **As a** user, **I want** to cancel my hotel reservations, **so that** I don't waste money on no-longer-needed reservations.

7) **As a** user, **I want** to be able to cancel my hotel reservations at any time, **so that** I can save on cancellation penalty.

8) **As a** Premium member, **I want** to cancel my hotel reservations at any time, **so that** I can save on cancellation penalty.

Which variant is correct?

Assume we need to specify a functionality "cancellation of hotel reservation", which should be available for all users of a booking site

1) **As a** user, **I want** to have a cancellation functionality.

2) **As a** user, **I want** to have a cancellation functionality, **so that** I can cancel my booking.

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7) **As a** user, **I want** to be able to cancel my hotel reservations at any time, **so that** I can save on cancellation penalty.

8) **As a** Premium member, **I want** to cancel my hotel reservations at any time, **so that** I can save on cancellation penalty.

Which variant is correct?

Assume we need to specify a functionality "cancellation of hotel reservation", which should be available for all users of a booking site

As a user, **I want** to cancel my hotel reservations, **so that** I don't waste money on no-longer-needed reservations.

OR

As a user, **I want** to be able to cancel my hotel reservations, **so that** I don't waste money on no-longer-needed reservations.

Which variant is correct?

Assume we need to specify a functionality "cancellation of hotel reservation", which should be available only for Premium members at any time – all other users have to pay cancellation penalty

1) **As a** Premium member, **I want** to have a cancellation functionality.

2) **As a** Premium member, **I want** to have a cancellation functionality, **so that** I can cancel my booking.

3) **As a** Premium member, **I want** to have a cancellation functionality, **so that** I can save on cancellation penalty.

4) **As a** Premium member, **I want** to have a "cancel booking" button, **so that** I can cancel my booking.

5) **As a** user, **I want** to cancel my hotel reservations, **so that** I don't waste money on no-longer-needed reservations.

6) **As a** user, **I want** to be able to cancel my hotel reservations at any time, **so that** I can save on cancellation penalty.

7) **As a** Premium member, **I want** to cancel my hotel reservations at any time, **so that** I can save on cancellation penalty.

Which variant is correct?

Assume we need to specify a functionality "cancellation of hotel reservation", which should be available only for Premium members at any time – all other users have to pay cancellation penalty

1) **As a** Premium member, **I want** to have a cancellation functionality.

2) **As a** Premium member, **I want** to have a cancellation functionality, **so that** I can cancel my booking.

3) **As a** Premium member, **I want** to have a cancellation functionality, **so that** I can save on cancellation penalty.

4) **As a** Premium member, **I want** to have a "cancel booking" button, **so that** I can cancel my booking.

5) **As a** user, **I want** to cancel my hotel reservations, **so that** I don't waste money on no-longer-needed reservations.

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PGRD Semester 1 2024 (2410)

Account

Dashboard

Courses

Home

Announcements

Syllabus

Modules

Discussions

Assignments

Grades

Week 2 - Project Charter vs. Vision & Scope Document

Week 2 - Overview & Your plan for this week

Week 2 - Lectorials and Workshops

SAMPLE: SE Challenge 1

User stories: Which variant is correct?

User stories: Which variant is correct?

Due No due date **Points** 10 **Questions** 10
Time Limit 10 Minutes **Allowed Attempts** Unlimited

Instructions

This is ungraded, unlimited attempt quiz for you to self-check your knowledge of user story format.

A detailed analysis of similar examples is discussed in Week 3 Lectorial.

Take the Quiz

Questions?